

From Vitushkin's Two-Dimensional Example to Fable's Three-Dimensional Jacobian Counterexample

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Abstract

In July 2026, a polynomial map $\mathbb{C}^3 \rightarrow \mathbb{C}^3$ with constant Jacobian determinant -2 was announced. The map is not injective, so it gives a counterexample to the Jacobian Conjecture in every dimension $n \geq 3$. Its construction is very close to an obscure 1999 Russian paper of Vitushkin. This note writes out, step by step, one way to derive the three-dimensional counterexample from Vitushkin's two-dimensional example, and shows that when the extra dimension degenerates, the construction falls back to the two-dimensional prototype.

Introduction

In July 2026, Levent Alpöge, Akhil Mathew, and Anthropic's Claude Fable 5 produced a three-dimensional counterexample to the Jacobian Conjecture: a polynomial map $\mathbb{C}^3 \rightarrow \mathbb{C}^3$ whose Jacobian determinant is the nonzero constant -2 , but sends three distinct points to the same image. Adjoining identity coordinates extends it to every $n \geq 3$, so the conjecture is now false in all dimensions ≥ 3 ; only the classical case $n = 2$ remains open.

The construction is very close to work of the Soviet/Russian mathematician Vitushkin in a 1999 Russian-language paper. That paper used σ -processes to build two-dimensional rational maps with constant Jacobian that are globally non-invertible, getting very close to the later three-dimensional example. Because it was written in Russian and was not widely read, the connection seems to have been missed.

This note has one purpose: to write out, step by step, the derivation chain from Vitushkin's two-dimensional framework to Fable's three-dimensional counterexample — from the two-dimensional fiber equation produced by the σ -process, up one degree to three dimensions, through a birational change of coordinates and a clearing of denominators, down to the three polynomials. It also shows that when the new dimension degenerates, the construction collapses back to the two-dimensional prototype. In this sense, the two examples look like two slices of the same construction, rather than two unrelated objects.

Starting point: Vitushkin's two-dimensional example

$$F = x^2y^6 + 2xy^2, \quad G = xy^3 + \frac{1}{y}, \quad J(F, G) = -2.$$

The obstruction is the $1/y$ in G : the Jacobian is a nonzero constant, and the map is globally non-invertible, but because of the denominator it is not a polynomial, so it is not yet a counterexample to the Jacobian Conjecture.

Step 1: reduce the map to a single (quadratic) equation

The basic point of a counterexample is that several inputs can land on the same output. So we ask the reverse question: fix an output; which inputs map to it? These preimages are exactly the roots of a certain equation in an auxiliary variable T — the *fiber equation* (the equation for the preimages of a fixed output). This equation is not arbitrary; it keeps track of how many preimages there are and where they sit.

How does it arise from Vitushkin’s original F, G ? Through a σ -process, a coordinate substitution designed to “look near infinity” (a blow-up type change) that spreads out the bad behavior crowded at infinity and clears denominators along the way. Vitushkin builds it from two pieces:

$$\Pi : x \mapsto \frac{x}{y}, y \mapsto \frac{1}{y}; \quad \Sigma_{xy} : x \mapsto xy, y \mapsto y.$$

His combination $\Pi \cdot \Sigma_{xy}^4$ amounts to the single substitution

$$x \mapsto xy^3, \quad y \mapsto \frac{1}{y}.$$

The powers of y it introduces cancel the high powers in F and the $1/y$ in G :

$$\begin{aligned} P &= F\left(xy^3, \frac{1}{y}\right) = (xy^3)^2 \left(\frac{1}{y}\right)^6 + 2(xy^3) \left(\frac{1}{y}\right)^2 = x^2 + 2xy, \\ Q &= G\left(xy^3, \frac{1}{y}\right) = (xy^3) \left(\frac{1}{y}\right)^3 + \frac{1}{1/y} = x + y, \end{aligned}$$

with $J(P, Q) = 2y$. The σ -process is not just an arbitrary algebraic manipulation. It is a choice of coordinate at infinity that exactly cancels the denominators and high powers, turning the rational pair F, G into the clean polynomial pair P, Q .

Fixing the output (P, Q) and solving for the preimages: from $Q = x + y$ we get $y = Q - x$, and substituting into P ,

$$P = x^2 + 2x(Q - x) = -x^2 + 2Qx \implies x^2 - 2Qx + P = 0.$$

A quadratic, two roots, hence 2-to-1. In the note’s normalized output variables a, b (set $a = P$, $b = 4Q$; the factor 4 on the second coordinate is the source of the “off by a factor of 4” in the literature), multiplying through by -2 gives the fiber equation

$$-2T^2 + bT - 2a = 0. \tag{1}$$

Here a, b are the outputs, T is the “label” of a preimage (namely the x above), and each root corresponds to a preimage. Vitushkin is stuck here: the denominator $1/y$ cannot be removed, and the degree of the equation (two) is already saturated, with no higher term available to “store” the pole.

Step 2: lift by adding a cubic term, carried by a new coordinate c

Add a cubic term cT^3 to the left of (1), raising the degree to three. Write this cubic polynomial as

$$\Phi(T) := cT^3 - 2T^2 + bT - 2a, \quad \Phi(T) = 0. \tag{2}$$

- $c = 0$: back to (1), two-dimensional, 2-to-1.
- $c \neq 0$: cubic, hence 3-to-1.

The new coefficient c is precisely the third output coordinate, carried by a new variable z . This is the step “2D $\rightarrow$ 3D.”

Step 3: tie the root to the pole

Designate a root of (2):

$$t = y + \frac{1}{x} = \frac{1 + xy}{x}.$$

Why this t ? Because it carries the same kind of pole as the stubborn $1/y$ in Vitushkin's example: the denominator $1/x$ is packed inside the root, carried by the root rather than exposed in the map's formulas.

Now we must control *branching*. The map is a three-sheeted cover (fix an output; the three preimages are the three roots of $\Phi(T) = 0$). Two preimages collide — the cover branches and the Jacobian degenerates — exactly when Φ has a double root, i.e. $\Phi(t) = 0$ and $\Phi'(t) = 0$. Thus $\Phi'(t)$ is the “where does it branch” detector; differentiating in T is the standard test for a repeated root. Force $\Phi'(t)$ to be a pure pole:

$$\Phi'(t) = 3ct^2 - 4t + b = \frac{2}{x}. \quad (3)$$

Then for finite x we have $\Phi'(t) = 2/x \neq 0$, so branching can only occur at $x = \infty$: the bad behavior has been pushed to infinity, and the finite part is a clean three-sheeted cover (which is what makes a constant nonzero Jacobian possible). This is the three-dimensional form of Vitushkin's “push the exceptional curves to infinity.”

Step 4: solve back for the three source coordinates (x, y, z)

The unknowns are the source coordinates (x, y, z) . The known quantities live on the cubic-model side: the root t , the pole r , and the new output c . This step matches the two coordinate systems by a three-dimensional birational change.

x . Set $r := \frac{2}{x}$ (the pole on the right of (3)), so $x = \frac{2}{r}$.

y . From the root $t = y + \frac{1}{x}$ and $\frac{1}{x} = \frac{r}{2}$, we get $y = t - \frac{r}{2}$.

z . With no ready-made equation, write z as an ansatz with undetermined coefficients:

$$z = Ar^2 + Btr + Dcr^3.$$

The coefficients are pinned uniquely by two requirements (all verified symbolically):

- *Jacobian identically -2 fixes D .* For this shape one computes $\det DF = \frac{1}{4D}$ regardless of A, B, D : it is already constant (independent of x, y, z). Setting it to the target -2 gives $D = -\frac{1}{8}$.
- *All three components polynomial fixes A, B .* With $D = -\frac{1}{8}$, the component c is automatically polynomial, but (see Step 5) b carries a $\frac{1}{x}$ pole and a a $\frac{1}{x^2}$ pole. Killing them gives $2A + B = 1$ and $12A + 8B = 3$, whose unique solution is $A = \frac{5}{4}$, $B = -\frac{3}{2}$.

Hence

$$z = \frac{5}{4}r^2 - \frac{3}{2}tr - \frac{c}{8}r^3.$$

The three relations

$$x = \frac{2}{r}, \quad y = t - \frac{r}{2}, \quad z = \frac{5}{4}r^2 - \frac{3}{2}tr - \frac{c}{8}r^3$$

are the three-dimensional birational change $(x, y, z) \leftrightarrow (t, r, c)$.

Step 5: substitute, clear denominators, obtain three polynomials

Write the outputs a, b, c in the source coordinates; each has a source.

b from (3). Solving $\Phi'(t) = 3ct^2 - 4t + b = \frac{2}{x}$,

$$b = 4t + \frac{2}{x} - 3ct^2.$$

$2a$ from (2). Solving $\Phi(t) = ct^3 - 2t^2 + bt - 2a = 0$,

$$2a = ct^3 - 2t^2 + bt.$$

c by inverting the z -formula. Solving $z = \frac{5}{4}r^2 - \frac{3}{2}tr - \frac{c}{8}r^3$ gives $c = \frac{10}{r} - \frac{12t}{r^2} - \frac{8z}{r^3}$; with $r = \frac{2}{x}$ (so $\frac{1}{r} = \frac{x}{2}$) this is $c = 5x - 3tx^2 - x^3z$, and then $t = y + \frac{1}{x}$ gives

$$c = 5x - 3\left(y + \frac{1}{x}\right)x^2 - x^3z = 2x - 3x^2y - x^3z.$$

All three still carry $t = \frac{1+xy}{x}$ and $\frac{1}{x}$. Substituting into $b, 2a$ and clearing denominators, the $\frac{1}{x}$ cancels (this is where the factor $(1+xy)$ appears), giving

$$\begin{aligned} a &= (1+xy)^3z + y^2(1+xy)(4+3xy), \\ b &= y + 3x(1+xy)^2z + 3xy^2(4+3xy), \\ c &= 2x - 3x^2y - x^3z. \end{aligned}$$

This is exactly Fable's three-dimensional example, with $\det = -2$.

The connection

$$\underbrace{-2T^2 + bT - 2a = 0}_{\text{Vitushkin, 2D, 2-to-1}} \xrightarrow{+cT^3 \text{ (new coord } c \text{ carries } z)} \underbrace{cT^3 - 2T^2 + bT - 2a = 0}_{\text{Fable, 3D, 3-to-1}}$$

Setting $c = 0$ in Fable's cubic collapses it exactly to Vitushkin's quadratic. Vitushkin's example is the $c = 0$ slice of Fable's; Fable's example is Vitushkin's equation lifted by a cubic term carried by z . This one-way derivation shows the two constructions are closely related.

The special fiber: seeing the 3-to-1

At the output $(a, b, c) = \left(-\frac{1}{4}, 0, 0\right)$ the cubic degenerates:

$$-2T^2 + \frac{1}{2} = 0 \implies T = \pm \frac{1}{2}.$$

- Two finite roots give the preimages $\left(1, -\frac{3}{2}, \frac{13}{2}\right)$ and $\left(-1, \frac{3}{2}, \frac{13}{2}\right)$.
- The third root goes to $T = \infty$ (i.e. $x = 0$), giving the preimage $\left(0, 0, -\frac{1}{4}\right)$.

Three preimages, matching 3-to-1.

This point lies exactly on the two-dimensional slice. Its third output coordinate is $c = 0$, and, as noted above, $c = 0$ collapses the three-dimensional equation to the two-dimensional quadratic. Taking the first two output coordinates $(a, b) = \left(-\frac{1}{4}, 0\right)$ down to two dimensions, the fiber equation is the same $-2T^2 + \frac{1}{2} = 0$ with the same roots $T = \pm\frac{1}{2}$. These are the two preimages of the two-dimensional example at this output (2-to-1), corresponding in the σ -model $P = x^2 + 2xy$, $Q = x + y$ to $(x, y) = \left(\frac{1}{2}, -\frac{1}{2}\right)$ and $\left(-\frac{1}{2}, \frac{1}{2}\right)$. In three dimensions the same two finite roots reappear, together with one root at infinity $T = \infty$ ($x = 0$, preimage $\left(0, 0, -\frac{1}{4}\right)$); this third sheet is what the extra dimension adds. The special fiber is a concrete instance of “3D = 2D plus one sheet at infinity”: the two finite sheets are already present in Vitushkin’s two-dimensional example, and the sheet at infinity is new to three dimensions.

Discussion

Laying out “who contributed what,” Vitushkin, the author of the 1999 Russian paper, seems to provide the larger part of the underlying mechanism, for concrete and checkable reasons.

1. The whole machine is his. The σ -process — the coordinate change for computing Jacobians and building examples at infinity — is the subject of his paper. The three-dimensional construction runs on it end to end.
2. The core local model is almost verbatim his. Vitushkin’s Example 1, after his own σ -process, gives $P = a^2 + 2ab$, $Q = a + b$ — exactly the local model of the $c = 0$ slice of Fable’s example, off only by a factor of 4 on the second coordinate. The engine producing “constant Jacobian yet globally non-invertible” is his, used almost unchanged.
3. The mechanism is his. “Constant nonzero Jacobian + globally many-to-one + the bad behavior hidden in a pole”: this entire geometric mechanism he demonstrated in two dimensions. He built the bad map; only the pole remained uncleared.
4. He located the obstruction. His Proposition 1 (a two-dimensional impossibility result) pins down exactly why two dimensions fail: the pole cannot be removed under these blow-ups. Knowing where the wall is tells you what to change — add a dimension. He marked both the obstruction and the direction of the fix.
5. He was one step from the finish. Method, a usable local model, the mechanism, the obstruction diagnosis — he had all four; what remained was “lift to three dimensions and balance the coefficients.”

What Fable adds is the last step that matters: choosing a new variable z to absorb the pole, tuning the coefficients so that all three components become polynomial with constant Jacobian, and actually turning the two-dimensional “impossibility” into a genuine three-dimensional counterexample. Anyone can write the framework; finding the specific balanced example is the hard part.

An important delimitation: saying Vitushkin has 60–70% refers to the composition of the mathematical contribution (machine, model, mechanism, and diagnosis are all his), not to a claim that Fable consciously copied him. This note gives a concrete derivation linking the two examples (the $c = 0$ collapse, the term-by-term match of local models), which is a checkable mathematical claim. Whether Fable actually retrieved and used the Russian paper is a question of provenance that mathematics cannot settle, and is not claimed here.

Conclusion

Returning to the opening question: are Vitushkin’s two-dimensional example and Fable’s three-dimensional counterexample related? Yes, and the relation is explicit. The whole chain is: use Vitushkin’s σ -process to read off the two-dimensional fiber equation (1); add a cubic term carried by a new variable z to lift it to the three-dimensional fiber equation (2); force the branch detector $\Phi'(t)$ to a pole (3), pushing branching to infinity; then solve back for the three-dimensional birational change and clear denominators to obtain the three polynomials a, b, c — Fable’s three-dimensional counterexample.

The chain is one-way, explicit, and verifiable: (1), (2), (3), the back-solving, and the clearing of denominators have all been checked symbolically, and the final map satisfies $\det = -2$, $\Phi(t) = 0$, and $\Phi'(t) = 2/x$. Setting $c = 0$ collapses the three-dimensional fiber equation exactly to Vitushkin’s two-dimensional equation, whose local model matches Vitushkin’s 1999 Example 1 term by term (off by a factor of 4 on the second coordinate). The two examples are more than lookalikes: one is the two-dimensional slice, and the other is the three-dimensional version of the same construction.

A few caveats. First, “why exactly this σ -chain” is taken from Vitushkin’s paper and quoted, not re-derived. Second, the specific birational change (especially the coefficients of z) is taken from the community’s reverse-engineering and may not be the original author’s form, with other equivalent forms possible. Third, this note proves mathematical homology and mutual derivability, but does not assert Fable’s actual provenance. In sum, this three-dimensional counterexample pushes Vitushkin’s twenty-year-old two-dimensional near-miss forward by one decisive step; the $n = 2$ Jacobian Conjecture remains open, and remains the original and most essential battleground of this two-dimensional machinery.